

Maag

Modular Agentic Autonomous Greenhouse

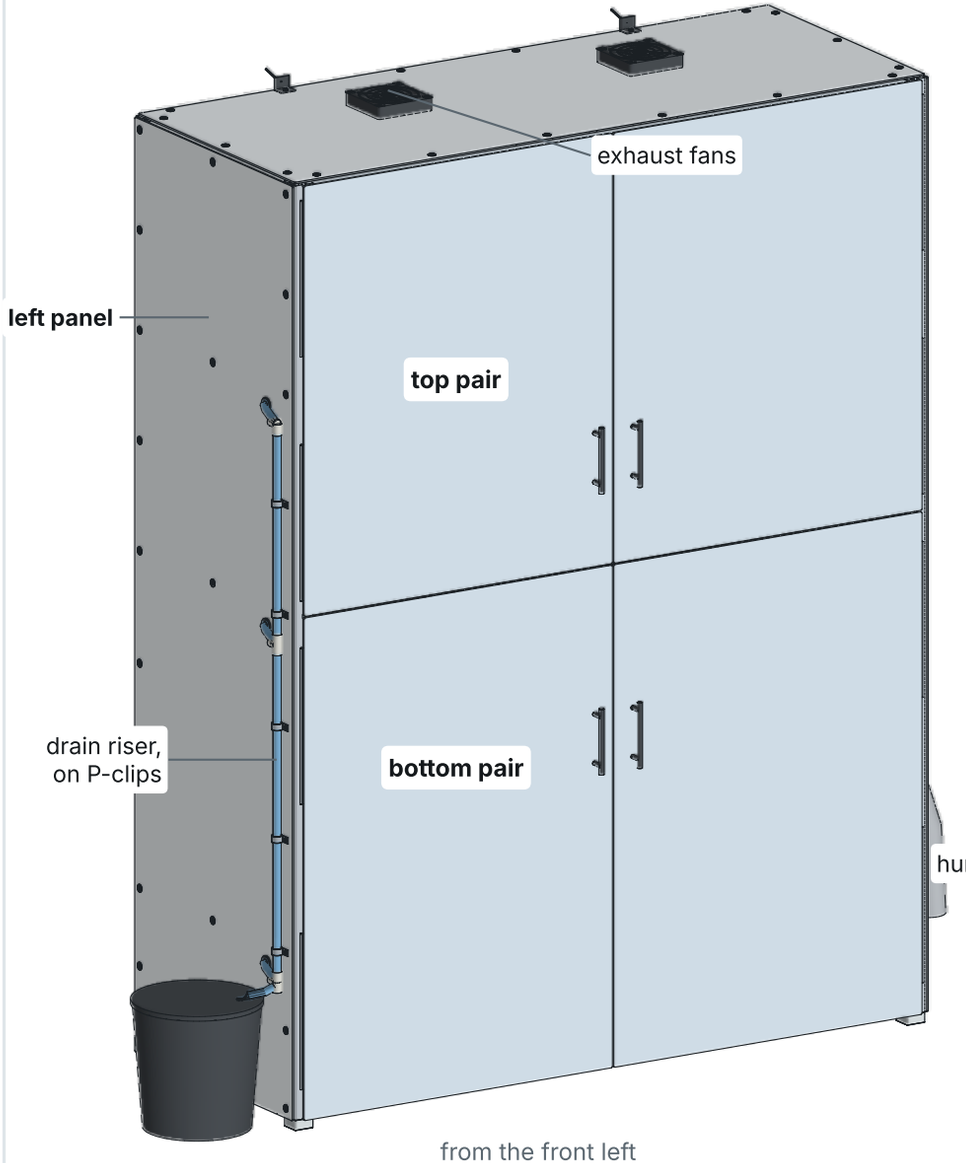
18 steps · tools · every hole

Enclosure assembly instructions, Enclosure Rev A. Frosted panels on the back, sides and top, four clear doors.

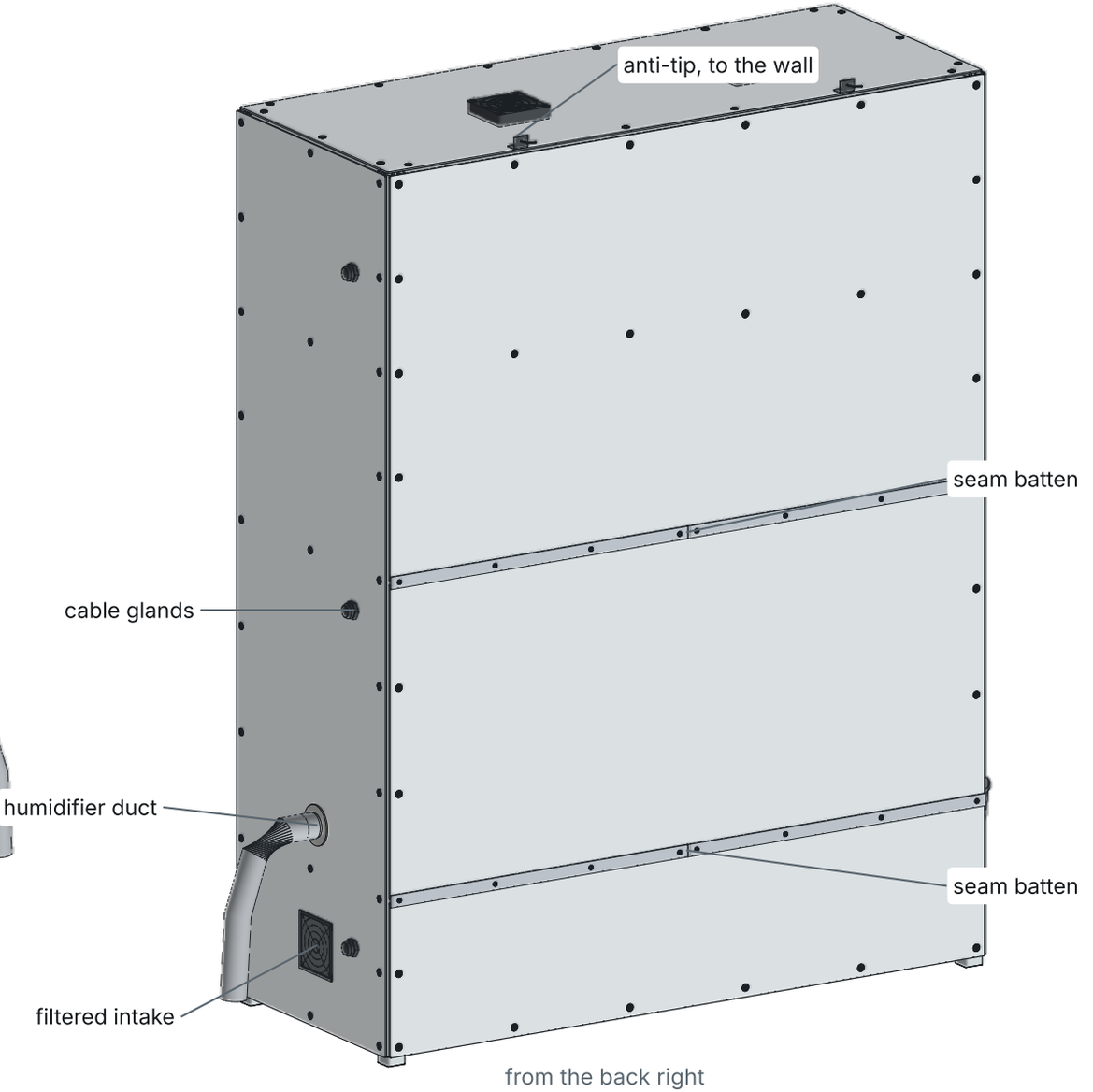
Design 0.5.0. The Maag cover render, the same on every booklet. Every drawing inside comes from enclosure.py and cabinet.py; the DXFs come from the same model.

The enclosure · what goes where

Six frosted panels, four clear doors. Everything surface mounted on the frame's outer faces.



from the front left

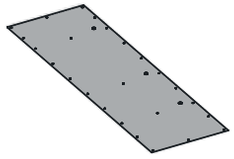


from the back right

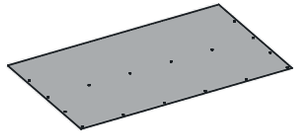
Back, sides and top: 1/8 in frosted polycarbonate (TUFFAK GP Matte), cut and drilled by A&C Plastics from the DXFs. Doors: 4.5 clear polycarbonate, SendCutSend. Every panel sits on EPDM foam tape beside each slot and is held by M5 screws into drop-in T-nuts, never more than 300 apart. Fit order: back, right, left, top, doors. The fans pull air out of the top; it comes in through the filter low on the right. The gantry is not drawn.

Enclosure · parts

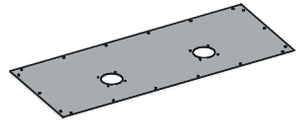
Count everything first. Quantities cover the complete enclosure.



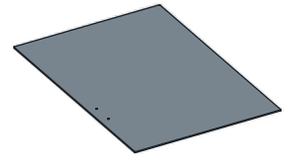
Side panels, frosted
2×



Back panels, frosted
3×



Top panel, frosted
1×



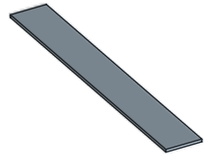
Doors, clear 4.5
4×



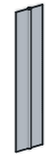
Seam batten, 5052
4×



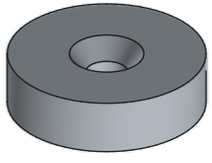
Hinge spacer, clear 3.0
8×



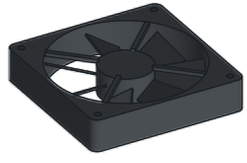
Astragal, clear 3.0
2×



Piano hinge, 1 ft
8×



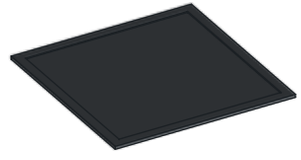
Cup magnet Ø16, M3
8×



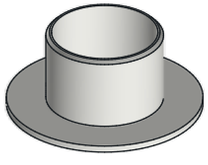
Exhaust fan NF-F12
2×



Fan grill NA-FG1-12
3×



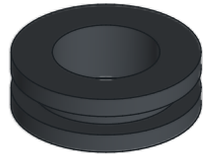
Dust filter 120
1×



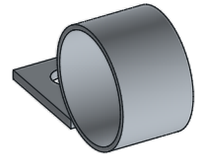
Duct collar, printed
1×



Cable gland PG21
3×



Grommet 7/8 in
3×



P-clip 5/8 in
5×

Also, no picture needed:

EPDM foam strip 3/16 × 1/2 in, 50 ft ×3 (28.6 m used)
EPDM weatherstrip 5/16 × 1/4 in, 17 ft ×2 (8.9 m used)
VHB 4905 tape, 1/2 in · steel discs Ø16 × 1, adhesive ×8
Door pulls, 96 centres ×4 · reed switches 7939WG ×4 (BOM G047)
Anti-tip kit, steel cable ×2 · neutral silicone (from the frame build)

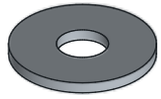
Enclosure · fasteners

Every screw in the build. Lay them out in labelled pots first.



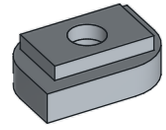
M5 × 16 button head · panels, clips

96×



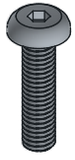
DIN 9021 washer M5 · 15 OD

89×



M5 drop-in T-nut, hammer head

112×



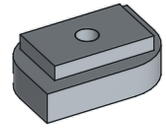
M5 × 20 button head · battens

16×



M3 × 12 flat head · hinges, magnets

40+×



M3 drop-in T-nut

40+×



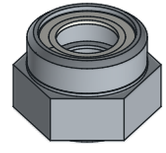
M4 × 40 button head · fans

8×



M4 × 16 button head · grill, collar

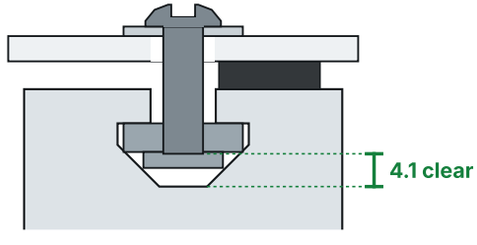
7×



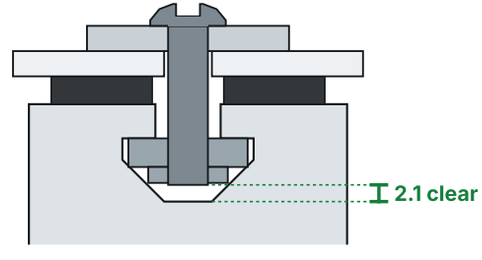
M4 nyloc + 2 washers

15×

Hinges: one M3 in every hole of the frame leaf (count them on yours; the drawings show four a hinge). Buy 50 M3 flat heads and 50 M3 T-nuts.
Pulls come with their own screws: use M4 × 10 instead, the door is only 4.5 thick. Drop-in T-nuts must be the kind that turns into the slot as you tighten.



a panel screw · M5 × 16

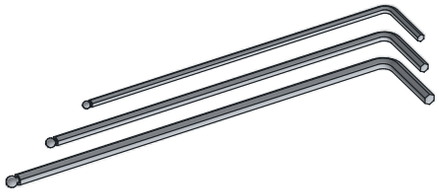


a seam · M5 × 20

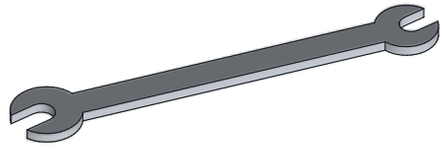
Tape squeezed from 4.8 to 3.5, panel 3.175, washer 1.2: an M5 × 16 reaches 8.1 below the face, 3.8 past the lips into the nut and 4.1 short of the slot floor. Never let a screw bottom: it feels tight and clamps nothing.

Tools

Everything the 18 steps need. Metric throughout; the tubing and tapes are sold in inches.



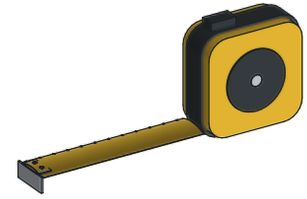
Hex keys, ball end
2 · 2.5 · 3 mm, long series
steps: 3: every M5, 4 to 13 · 2.5: M4, 7, 12, 14
2: the M3 flat heads, 15, 16



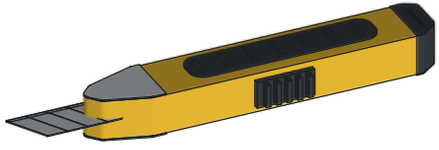
Spanner
7 mm
steps: the M4 nylocs inside: fans,
grill, collar, 7 and 12



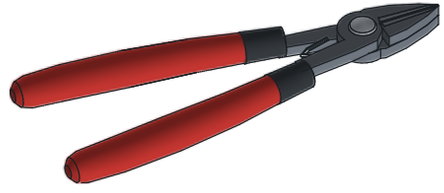
Calipers
150 mm, digital
steps: the hinge's hole line, 1;
the centre gap, 17



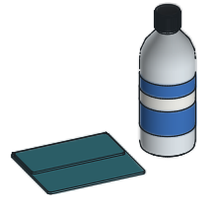
Tape measure
5 m
steps: T-nut positions off the face maps,
3 to 13



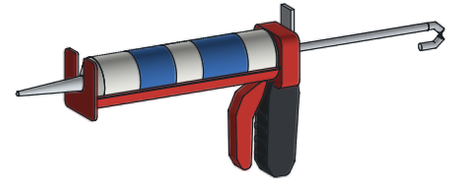
Utility knife
18 mm snap-off, and scissors
steps: tape and weatherstrip ends,
3, 8, 10, 13, 15



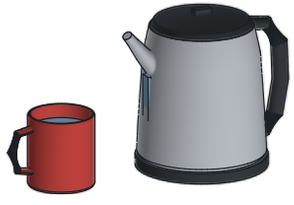
Side cutters
125 mm, flush cut
steps: the riser's zip ties, 9



Isopropyl alcohol
and microfibre cloths
steps: every land before tape, every
bond before VHB, 2 and 14



Caulking gun
and neutral-cure silicone
steps: the post tops, 2



Hot water
a kettle and a mug
steps: softening the drain tube
ends, 9 and 11

And a step ladder for the top, a soft surface to lay panels on, painter's tape for labels, and a helper for every panel and door: they are large, light and flex.

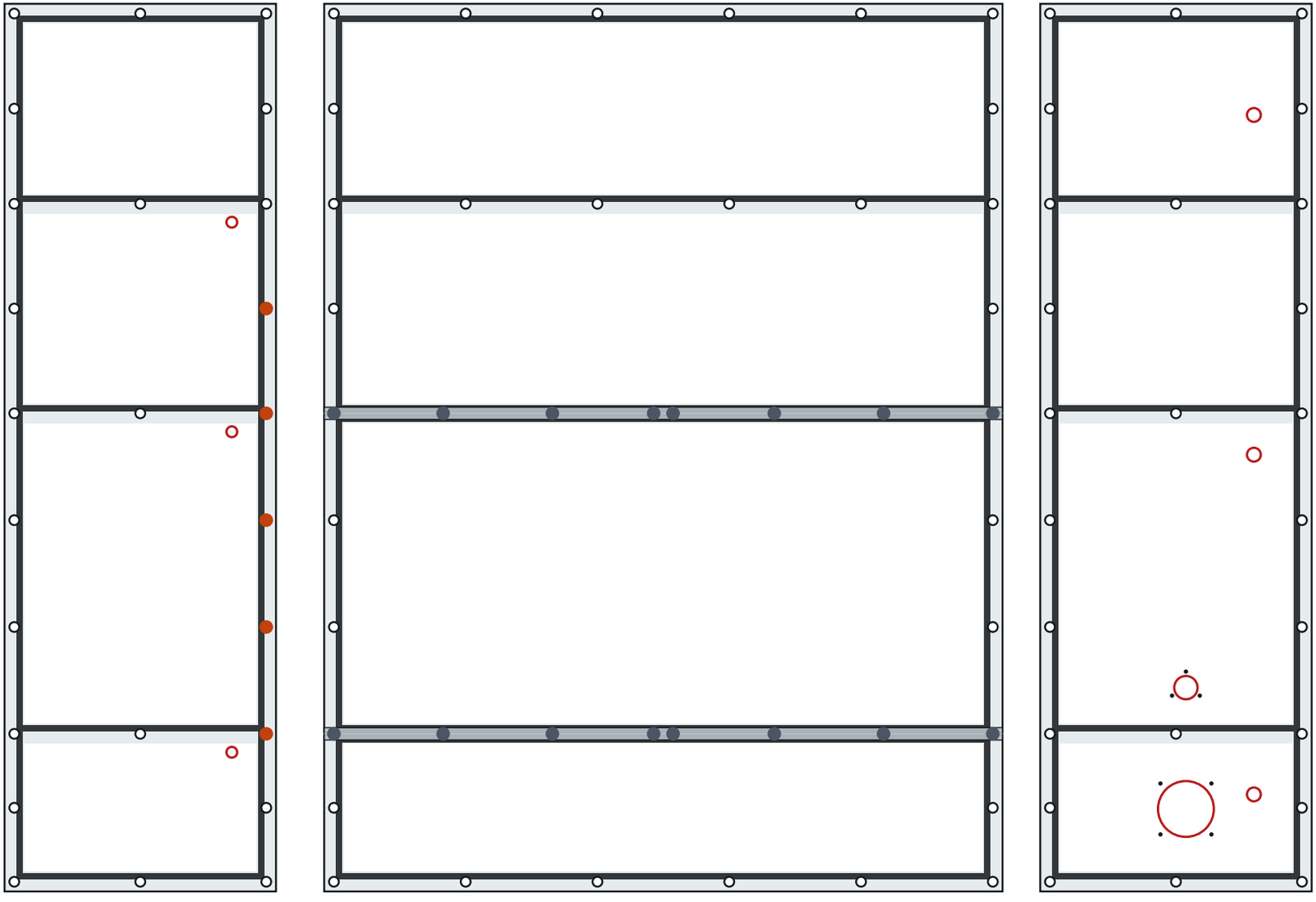
Where every screw goes · sides and back

Each face as you see it from outside. Drop a T-nut in the slot under every circle.

LEFT
front edge on the right

BACK
seen from behind

RIGHT
front edge on the left



- M5 × 16 + DIN 9021 washer
- M5 × 20, batten
- M5 × 16, P-clip
- M5 × 16, anti-tip bracket
- EPDM tape, 3/16 × 1/2 in
- frame behind
- cutout

T-nuts: left 25, right 25, back 44. The P-clips (orange) are panel screws too: they go in at step 11.

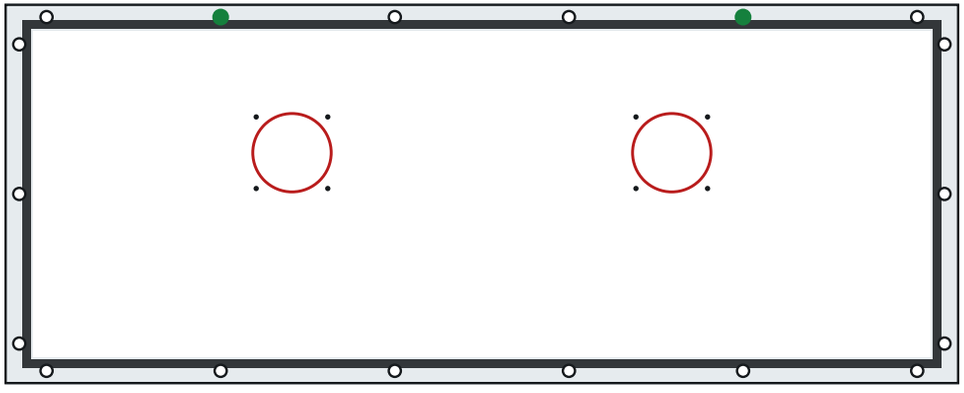
Every circle is on a slot centreline, 20 in from an edge or on a rail's middle. Seams: the back's three panels meet on B1 and B2 with a 6 gap; the batten screws pass through the gap. Cutouts: glands, intake and duct on the right; the three drain grommets on the left, 18 under each tier's rail.

Where every screw goes · top and front

The top seen from above, front edge down. The front: what the doors close on.

TOP

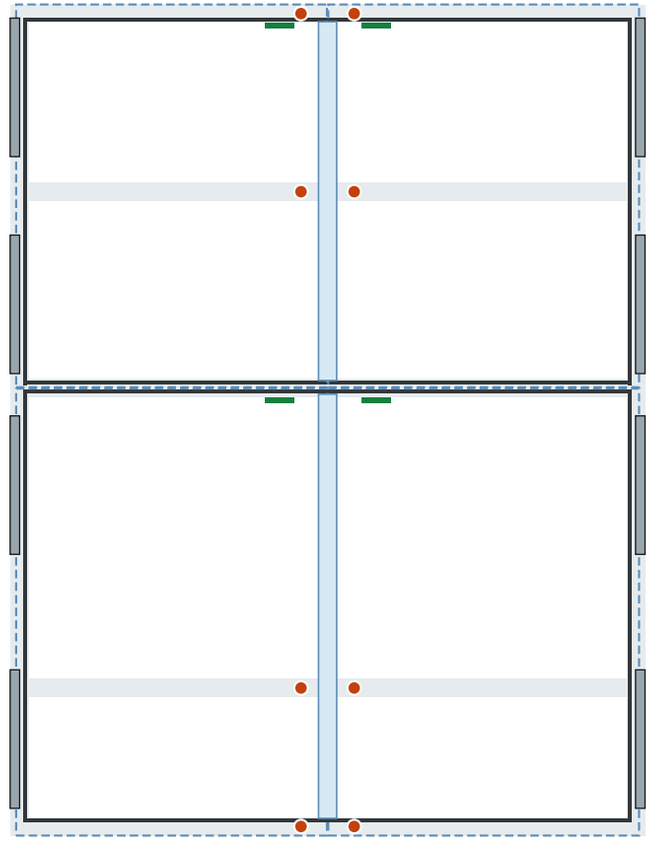
front edge at the bottom



- M5 × 16 + DIN 9021 washer
- M5 × 20, batten
- M5 × 16, P-clip
- M5 × 16, anti-tip bracket
- EPDM tape, 3/16 × 1/2 in
- frame behind
- cutout

FRONT

the frame face, doors dashed



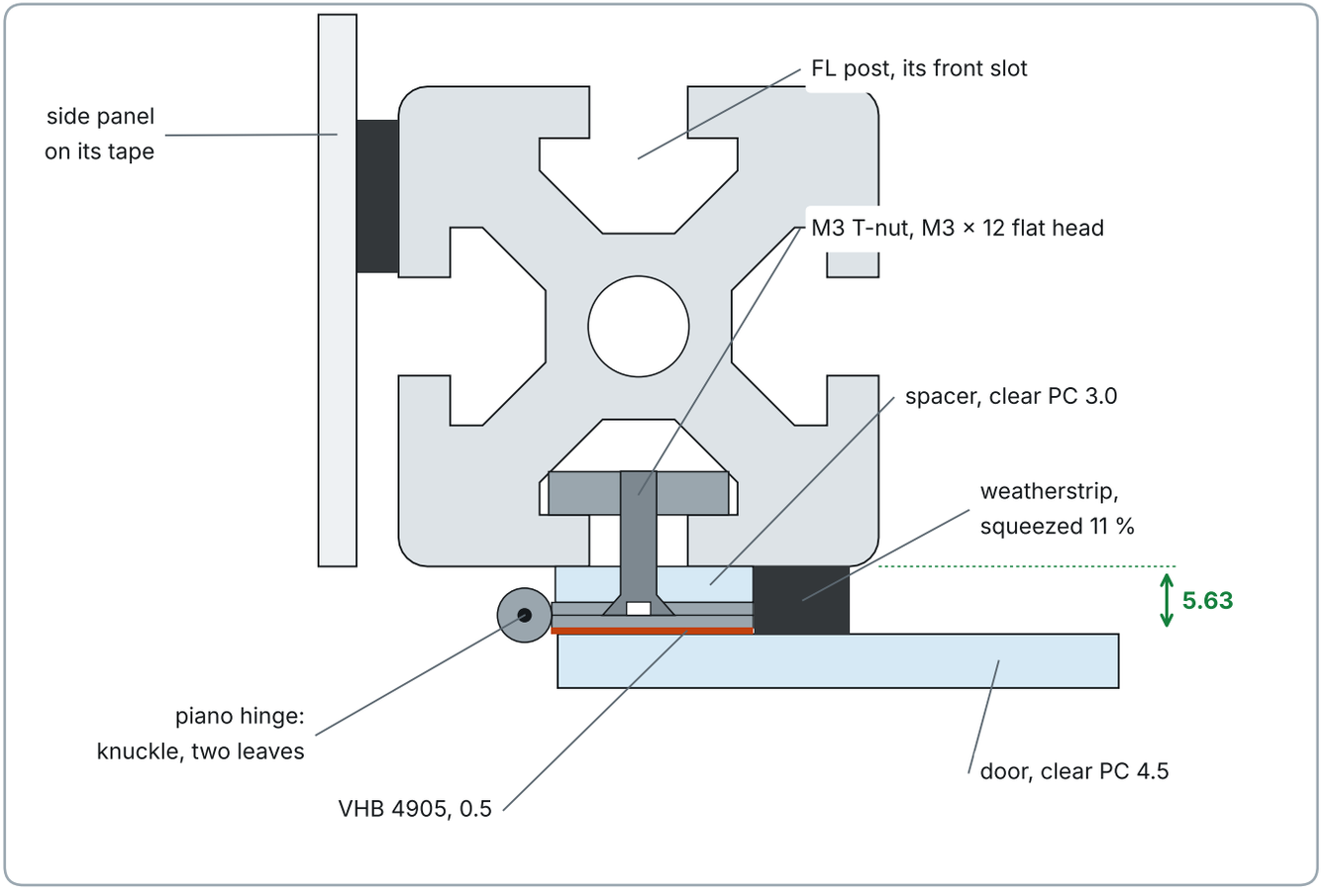
- hinge, frame leaf on its spacer
- cup magnet, on the slot line
- weatherstrip 5/16 × 1/4 in
- reed switch, under the rail
- astragal, on the right door

Bottom pair: magnets on F0 and F1, switch under F2. Top pair: magnets on F3 and F4, switch under F4. F2 carries the split, so its slot holds nothing.

Top: 18 T-nuts, none on the post ends (no slot there); the two green ones hold the anti-tip brackets. Fans at x 420 and 977, 340 back from the front: over the corridor, clear of the gantry's mast. Front: the doors meet 3 apart at the middle; the right door's astragal covers the gap.

How it closes · the door, in section

True scale, looking down on the bottom left door's hinge. Every number is the model's.



Hinge stack

spacer 3.0 + leaves 2 × 1.07 + VHB 0.5 = 5.63

Catch

cup magnet 4.5 + steel disc 1.0: 0.13 of air

Seal

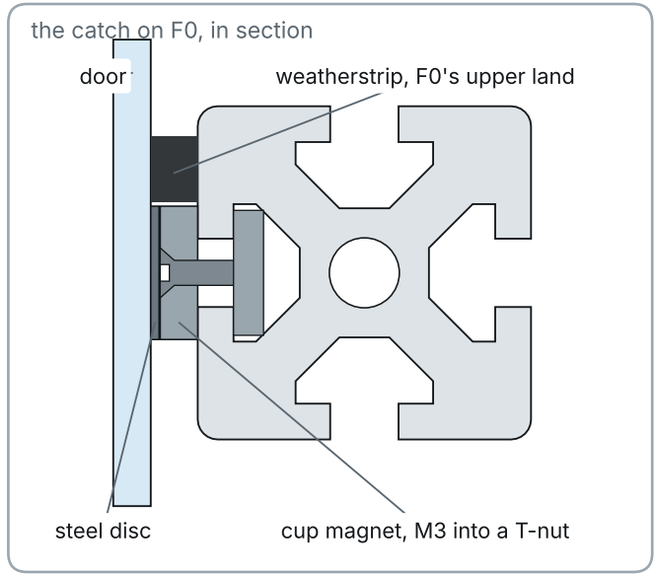
weatherstrip 6.35 to 5.63: light,
so the magnets win

Astragal

2.13 clear of the rails,
0.5 under the left door

Swing

152° before the door meets the post



The hinge's leaf holes sit 9.5 from its pin, so with the pin 10.5 in from the post's outer face the screws land on the slot line. If yours measure other than 9.5 (step 1), the spacer's slot still takes them from 8 to 11; the door then sits that much further in or out, and the centre gap changes by twice the difference. Open the left door of a pair first: its edge is over the right door's astragal.

1 Before you start

The cabinet finished to the frame and gantry booklets. Five checks, then label everything.

- [ ] **Frame Rev A done**
  Step 21 of the frame booklet signed off: square, plumb, drains water-tested.

- [ ] **Gantry in**
  Gantry Rev E finished and homed. The panels close it in: no reaching in from the back after step 5.

- [ ] **Parts counted**
  Six frosted panels, four doors, eight spacers, two astragals, four battens: against the parts page.

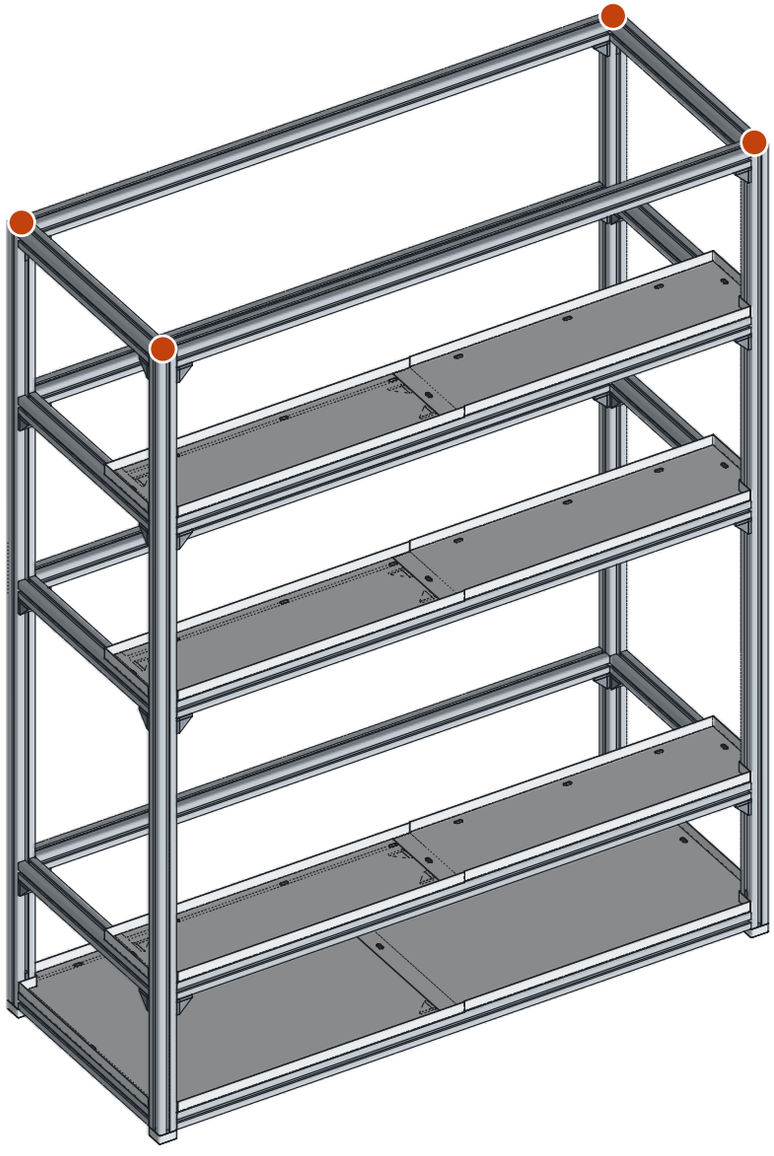
- [ ] **Hinge measured**
  Calipers from the pin to the centre of a hole: 9.5 expected, 8 to 11 works (page 'How it closes').

- [ ] **Panels labelled**
  Painter's tape on each, its name and OUT on the outer face. Keep the film on until it is on the cabinet.

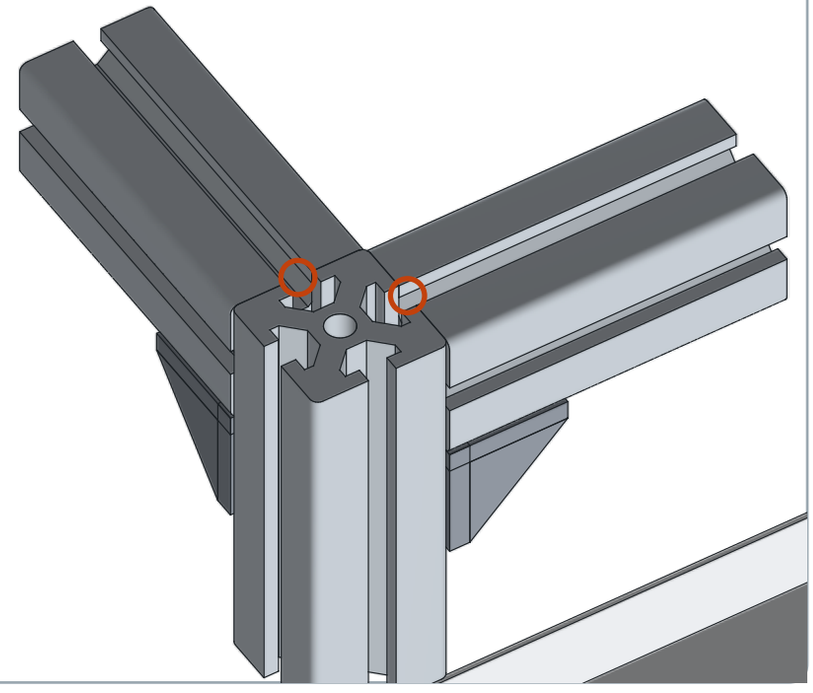
Work in this order: back (from the inside out, while you can still reach round it), right, left, top, then the doors. Polycarbonate scratches easily and flexes: lay panels on a blanket, carry them on edge, two people for anything over 1 m.

2 Clean every land, seal the post tops

Tape and VHB stick to clean aluminium only. The post tops are the one leak the tape misses.



FL from above: fill the two inward slot ends



Isopropyl alcohol, lint-free cloth

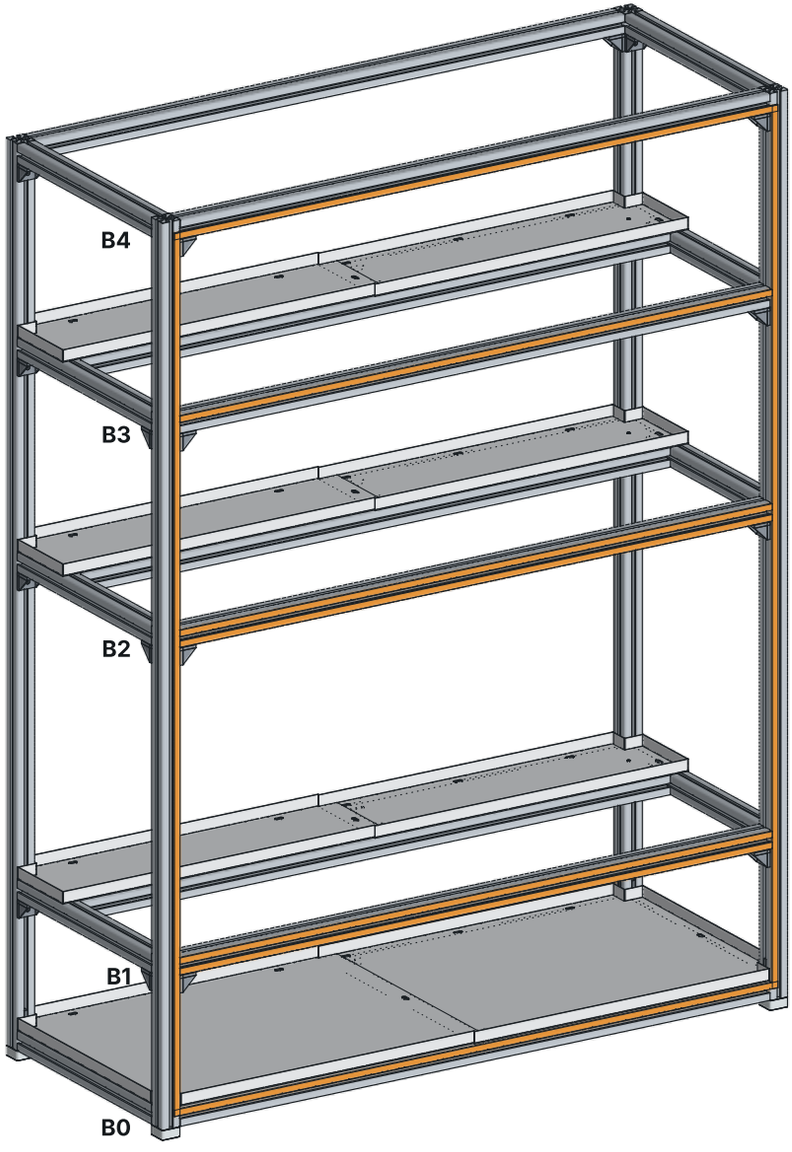
8× Neutral-cure silicone, a dab at each post top

caulking gun

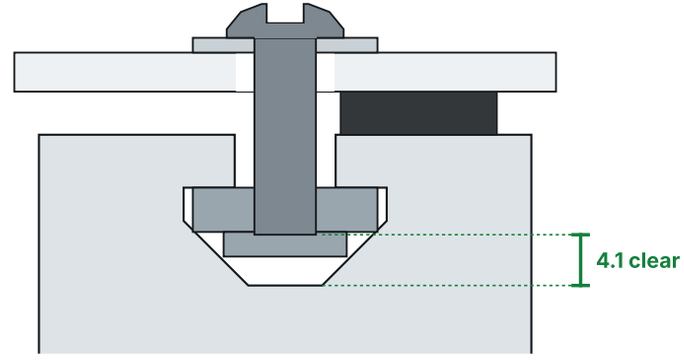
Wipe the outer face of every post and outer rail, both lands either side of each slot, with alcohol until the cloth comes away clean. At each post top, push a dab of silicone into the two slot ends that face into the cabinet and strike it flush with the end: under the top panel those slot ends would otherwise let air round the tape. Let it skin over (an hour) before step 13.

3 Back · tape

EPDM strip on the land beside every slot, on the side toward the bay; both lands on B1 and B2.



one rail, in section, once the panel is on



strip 12.7 wide, 4.5 off the slot line,
squeezed from 4.8 to 3.5

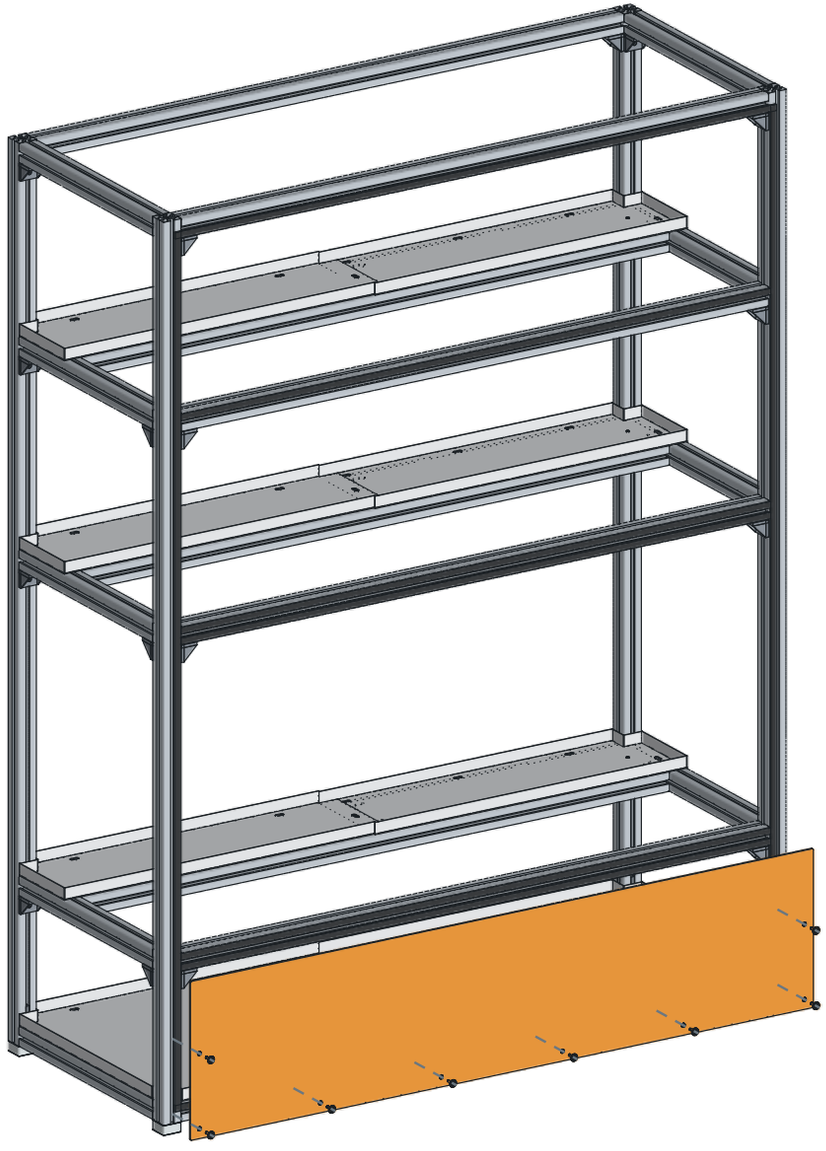
EPDM foam strip 3/16 × 1/2 in

`utility knife · tape measure`

Posts first, full height on the land toward the middle; then each rail between them. Lay it straight off the roll with the liner still on its back, then peel and press as you go: never stretch it, stretched foam creeps back and opens a gap. B0 and B4 on their inner land, B3 on its upper land, B1 and B2 on both: a panel edge lands on each. Butt the ends, do not overlap them.

4 Back · the bottom panel

T-nuts on their screws, the panel on the tape, every screw started, then tightened.



- 1× Back bottom panel · 1397 × 321.8
- 8× M5 × 16 + DIN 9021 washer
- 8× M5 drop-in T-nut

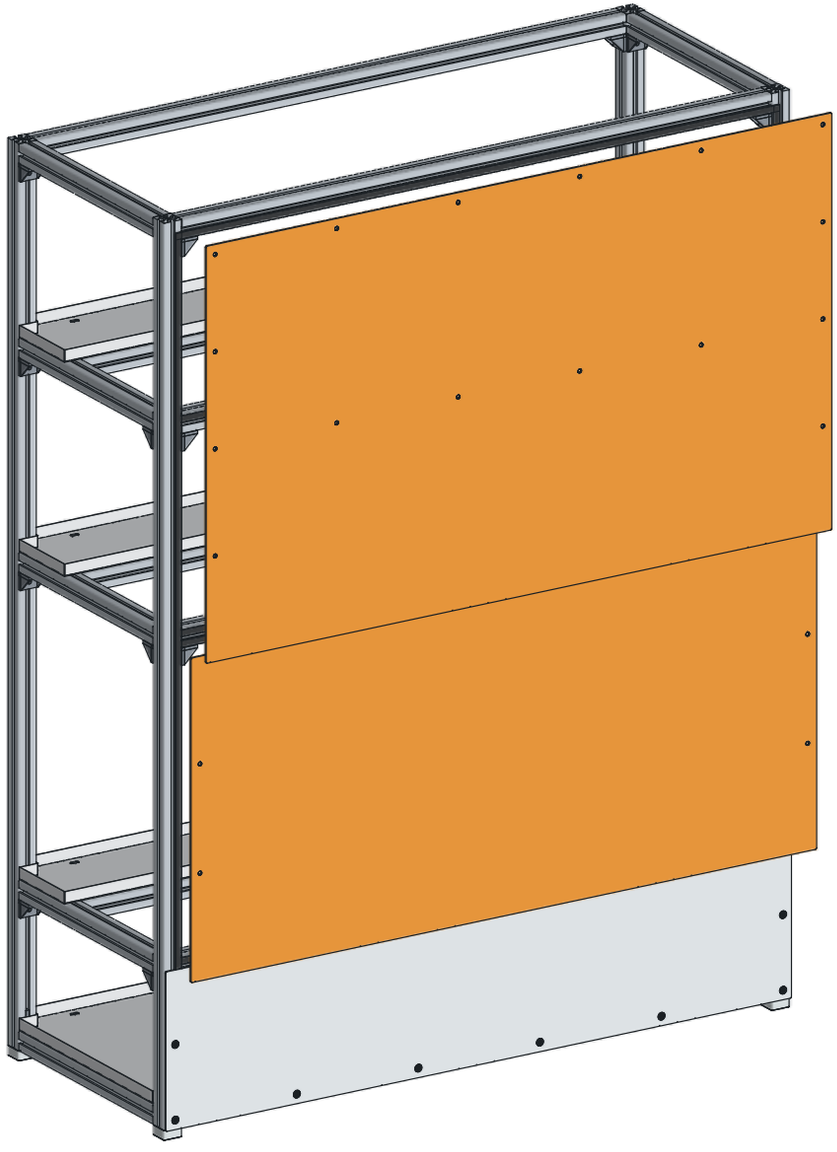
3 mm hex

The panel's top edge stops 3 short of B1's slot line: the seam gap. Its holes are 20 in from the sides and the floor edge.

Peel the film off the inner face only. Put each screw through a washer and the panel from outside and spin a T-nut two turns onto its tip. Offer the panel to the frame, push each nut into its slot, and turn each screw until the nut swings across and grips. Centre each screw in its 8 hole (the panel grows 1.89 across for 20 °C), then tighten from the middle out until the tape is about 3.5: a 3 mm hex key slides under the panel's edge, a 4 does not.

5 Back · the middle and upper panels

The same, twice. Each lands on its own land of B1 or B2; the 6 gap stays open for the battens.



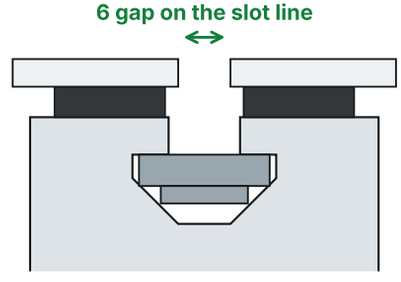
1× Back middle panel · 1397 × 654.4

1× Back upper panel · 1397 × 840.8

20× M5 × 16 + washer, T-nut

3 mm hex

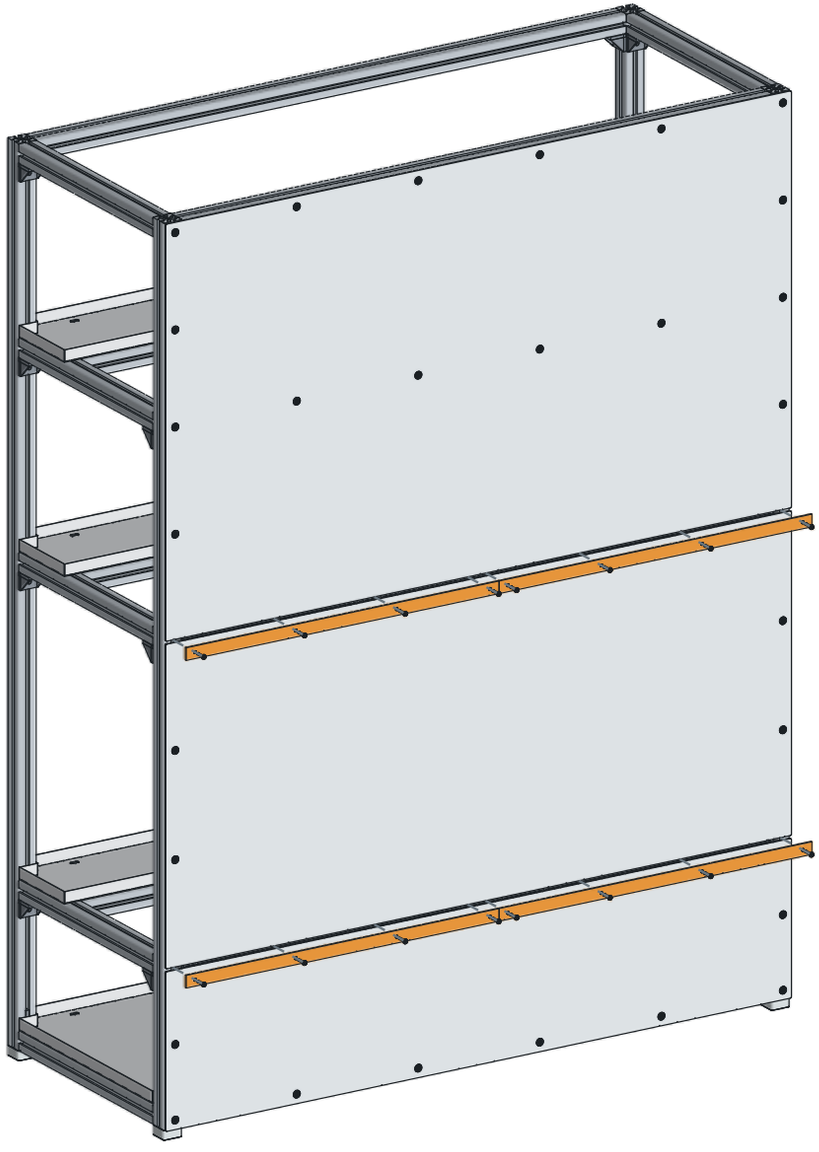
the seam on B1, before its batten



Middle panel next: its bottom edge 3 above B1's slot line, its top edge 3 below B2's, four screws into the posts only; hold it with two short strips of painter's tape across each seam while you start them. The upper panel the same, from B2's line to the top. Leave the seam screws out: they go through the battens, next.

6 Back · two battens over the seams

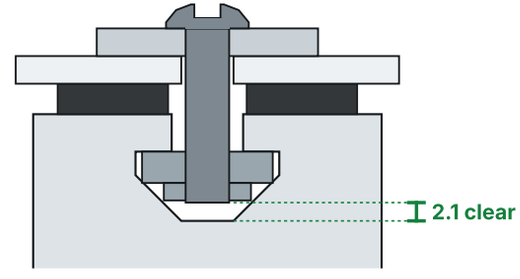
Two pieces a seam, butted at the middle. Eight M5 × 20 through each batten and the gap.



- 4× Seam batten, 698 × 25.4 × 3.2
- 16× M5 × 20 button head
- 16× M5 drop-in T-nut

3 mm hex

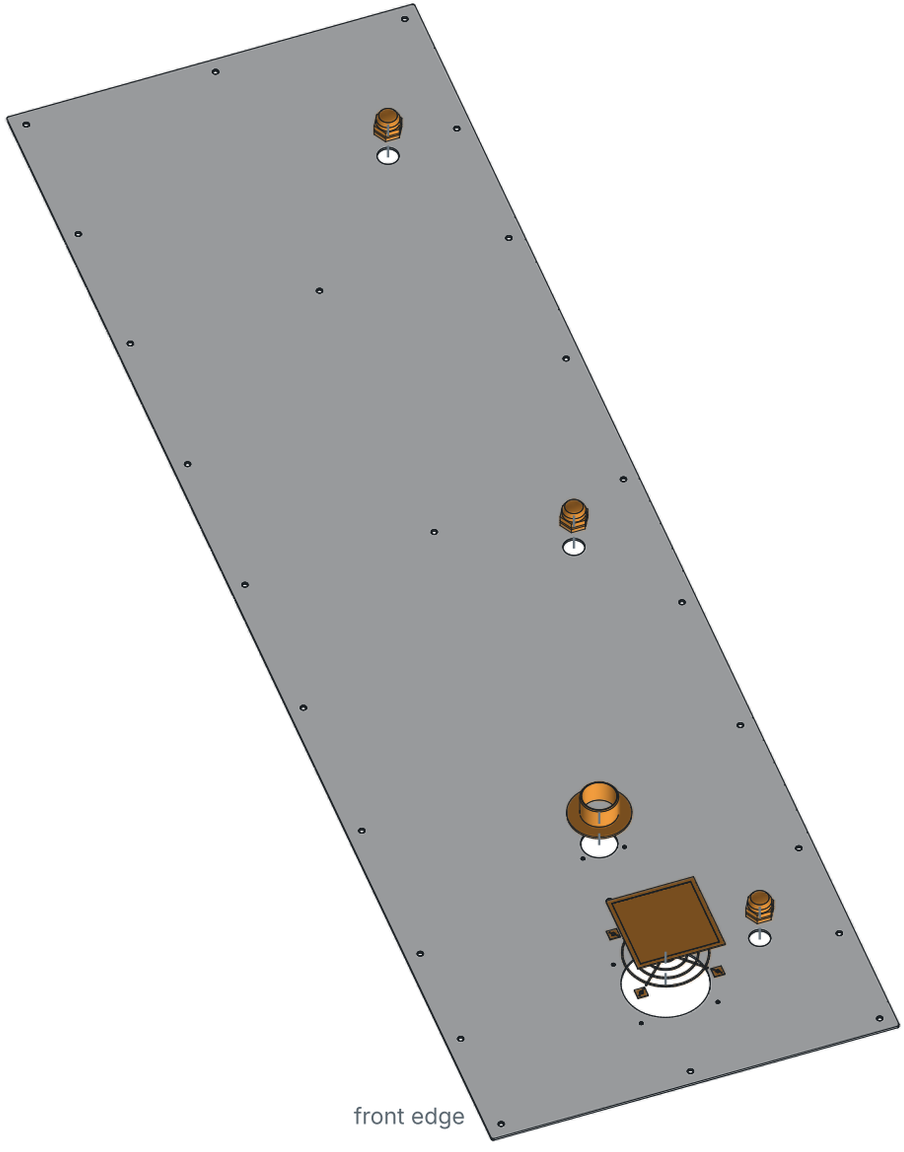
a seam, in section



The four battens are the same part: the right-hand one of each seam is a left-hand one turned end for end. Hole 1 sits 20 from its outer end. Each screw passes through the 6 gap between the panels into a T-nut on the rail's slot line (on the posts at the ends). Snug only: the batten pulls both edges onto their tape.

7 Right panel · on the bench first

Three glands, the intake grill and filter, the duct collar: easier flat than standing.



front edge

- 3× Cable gland PG21, nylon, its locknut inside
- 1× Fan grill NA-FG1-12, outside
- 1× Magnetic dust filter 120, on the grill
- 1× Duct collar, PETG, printed
- 7× M4 × 16 + washers + nyloc

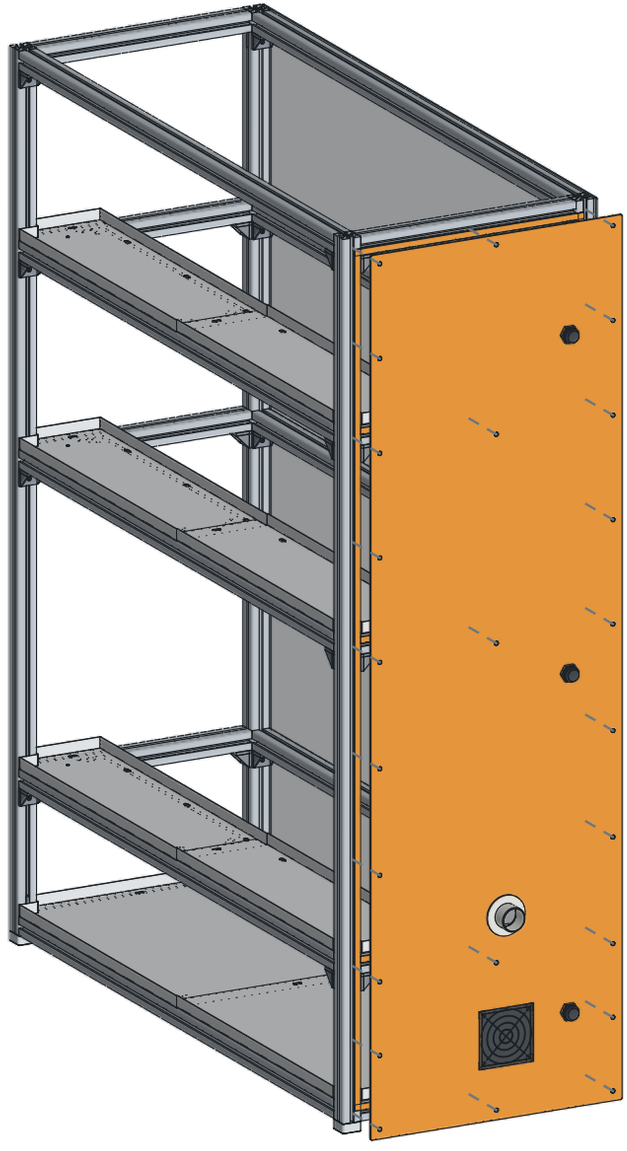
2.5 mm hex · 7 mm spanner

Print the collar from print/duct_collar.stl: PETG, 4 walls, 30 % infill, flange down.

Lay the panel outer face up (OUT on your label). Glands: body from outside, locknut inside, hand tight plus a quarter. Grill outside over the intake opening, four M4 × 16 with a washer each side and the nyloc inside; the filter's magnetic frame sticks to the steel grill. Collar flange outside, three M4 × 16. The 2 in duct slides over the spigot later and takes a zip tie.

8 Right panel · tape, then the panel

The same land pattern as the back: posts full height, rails between, inner lands at top and floor.



EPDM foam strip, right face

1× Right panel, with its glands and intake

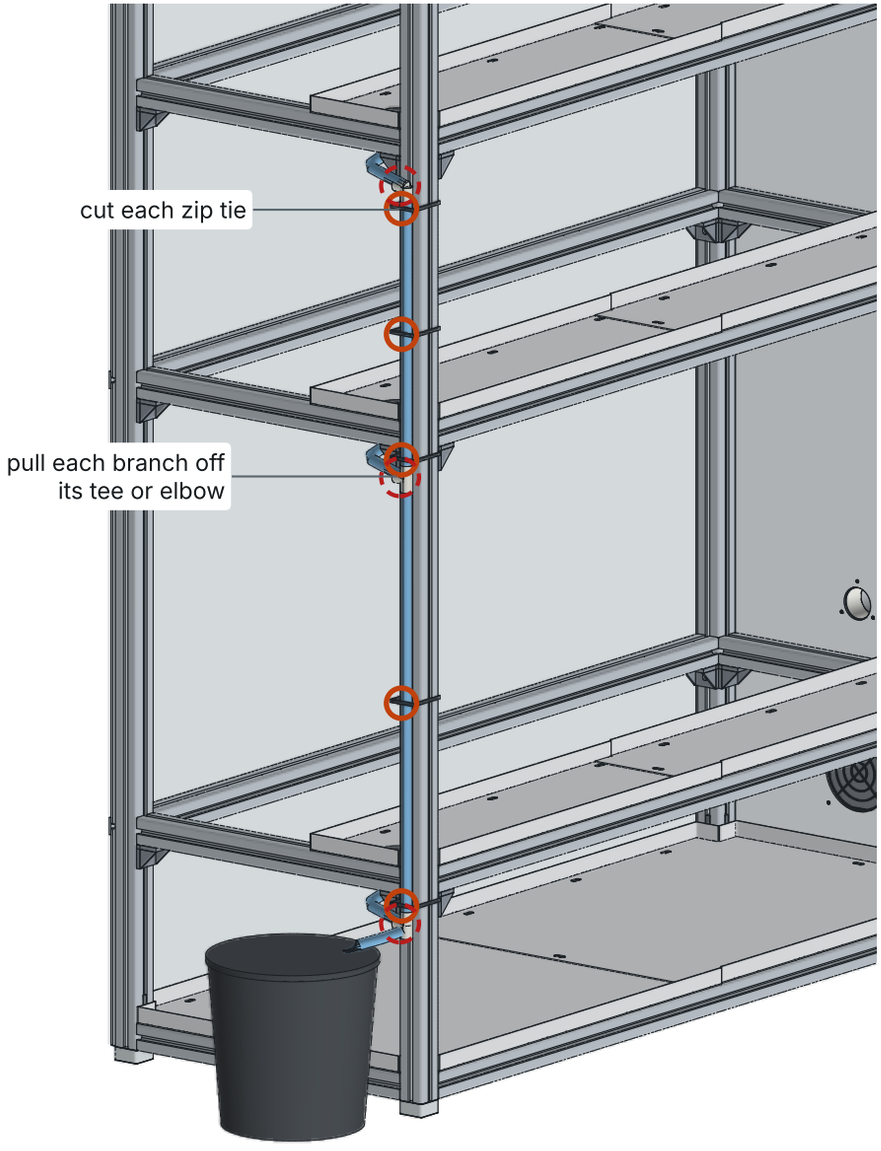
25× M5 × 16 + washer, T-nut

3 mm hex · utility knife

Tape as at step 3, on the right face. Then the panel: the front edge (glands at the back) flush with the front of the cabinet, screws and nuts as at step 4. Feed the duct over the collar's spigot and zip-tie it; its other end goes over the humidifier's outlet, standing on the floor to the right. Cables come through the glands later: tighten the dome nuts only then.

9 Left side · take the riser off the post

The riser moves outside the panel. Cut its five ties, pull the three branches off their fittings.



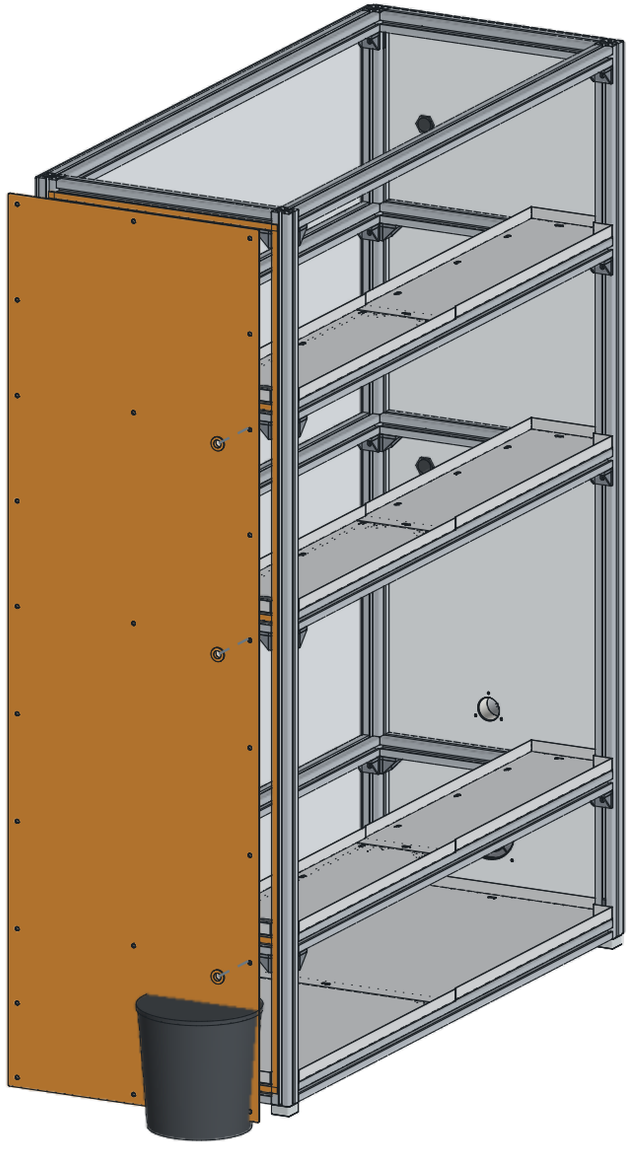


side cutters · a mug of hot water

Cut the five ties round FL and the riser. Dip each branch's end in hot water for half a minute and pull it straight off its stub: never lever it, a split barb leaks. Lift the riser, with its elbow, tees and the bucket line, off the post and lay it on the floor by the bucket. The branches stay on their bulkheads; leave their ends hanging out of the side.

10 Left side · tape, grommets, the panel

Grommets into the panel first, then the panel on, each branch end through its grommet.



EPDM foam strip, left face

3× Grommet, 7/8 in hole, 5/8 in bore

1× Left panel

20× M5 × 16 + washer, T-nut

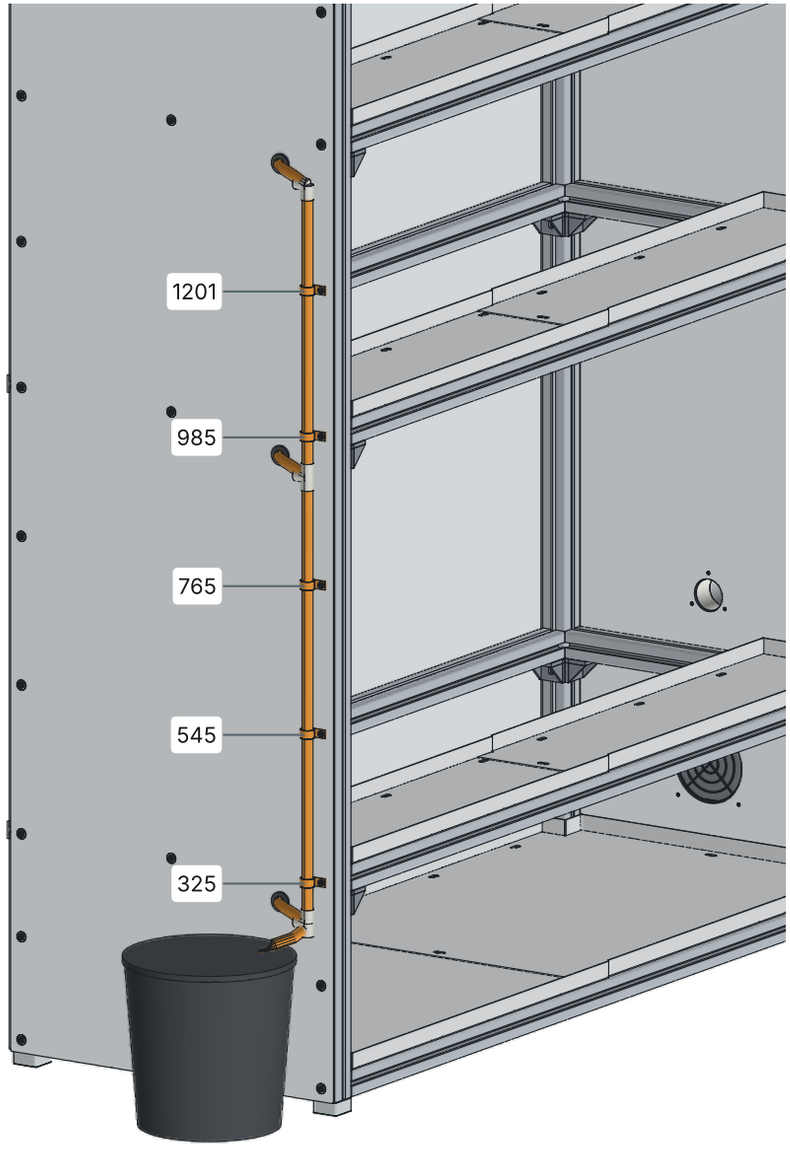
3 mm hex

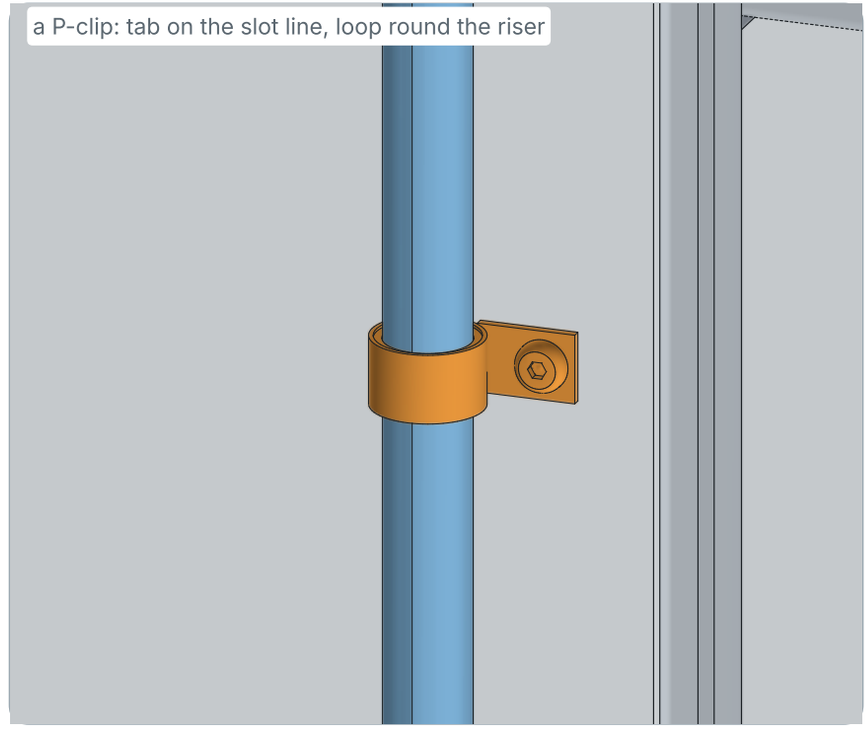
Leave out the five screws on FL's line from 325 to 1201: they are the P-clips' (step 11). Drop their T-nuts in now.

Tape the left face as before. Press the three grommets into the panel. With a helper holding the panel off the frame, feed each branch's end out through its grommet from the inside (A lowest, C highest), then bring the panel onto the tape and screw it as at step 4. A little dish soap on the tube makes it slide.

11 Left side · the riser on its P-clips

Five cushioned clips on FL's line hold the riser 2 off the panel. Trim each branch 8, refit.




a P-clip: tab on the slot line, loop round the riser

5× P-clip 5/8 in, EPDM cushioned

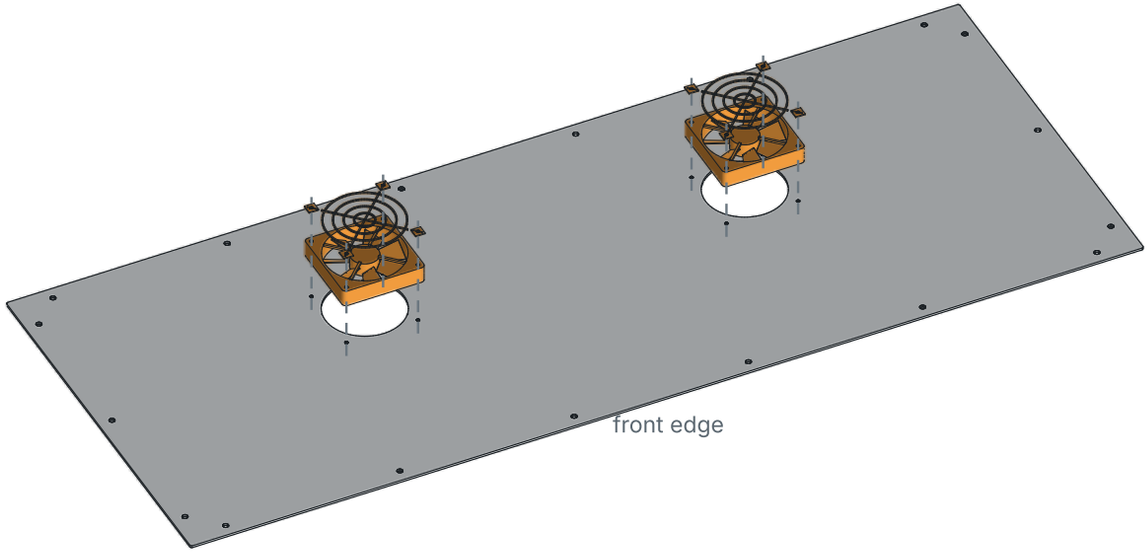
5× M5 × 16, no washer, into its T-nut

3 mm hex · hot water · knife

Clip the riser on at the five heights, its tees facing back toward the grommets. Each branch now has a shorter run: cut 8 off its end, square, warm it, and push it fully onto its stub. Check the fall: from every grommet down to the riser, never up. The riser stays clear of the panel (vinyl against polycarbonate clouds it over time) and drops into the bucket lid as before.

12 Top panel · fans on the bench

Two exhaust fans on the outer face, blowing up and out, a grill over each.



- 2× Noctua NF-F12 PWM, label side down
- 2× Fan grill NA-FG1-12
- 8× M4 × 40 + washers + nyloc

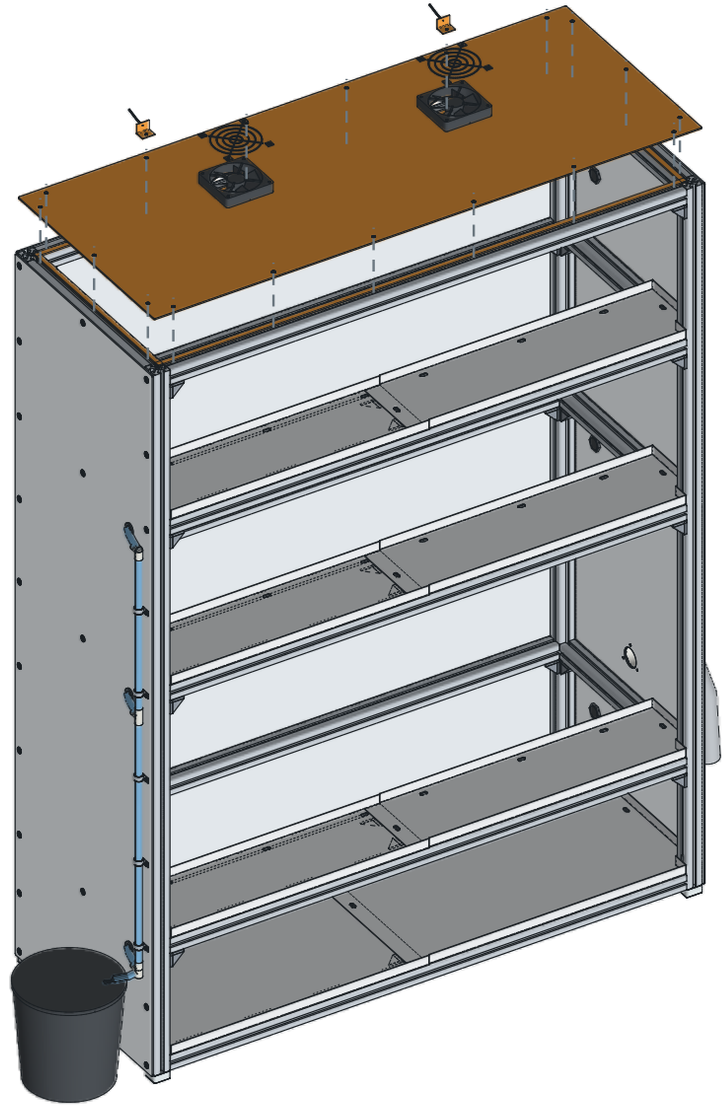
2.5 mm hex · 7 mm spanner

Air leaves upward: the fan's label (the outlet side's struts) faces the grill. Lead each cable to the back right, into the cabinet through the nearest gland.

Through-bolt each fan: screw, washer, grill tab, fan's corner flange, the panel, washer and nyloc on the inside. Snug, not tight: the fan's frame is plastic. Both fans run off the tier C node's exhaust output (BOM G022), in parallel. The fans sit outside so nothing new hangs into the gantry's corridor.

13 Top panel · on, and anchored to the wall

Tape on F4, B4, L4 and R4, the panel on it, then two anti-tip brackets into wall studs.



EPDM foam strip, top face

1× Top panel, fans on

16× M5 × 16 + washer, T-nut

2× Anti-tip kit: bracket, cable, wall bracket

2× M5 × 16 into B4, through the bracket

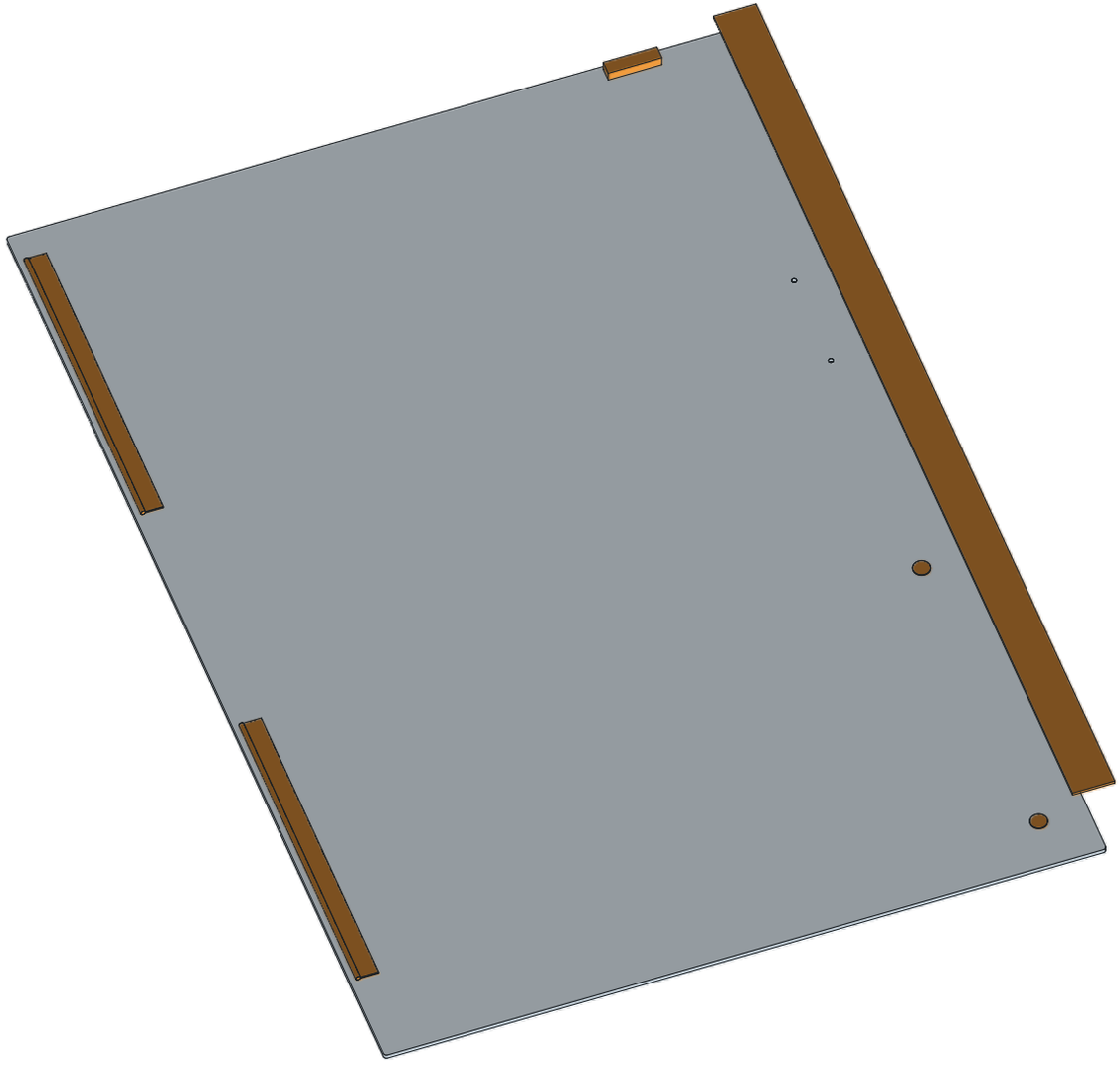
3 mm hex · step ladder · stud finder

The silicone at the post tops (step 2) must have skinned. No screw goes into a post end: the corner screws sit 60 in, on the rails.

The top's strips run on each top rail's inner land. Lift the panel up with its front edge to the front (the fans nearer the front than the back), centre it, screw it as at step 4. The two brackets go on B4's line at x 315 and 1082, in place of two panel screws; their cables run to wall brackets screwed into studs, pulled taut. The cabinet is 1.85 m tall and only 0.56 deep: this is not optional.

14 Doors · on the bench

Each door gets two hinges, two steel discs, a pull and a reed magnet; each right door its astragal.



- 8× Piano hinge 1 ft, door leaf on VHB
- 8× Steel disc Ø16, adhesive
- 4× Reed magnet, its own adhesive
- 2× Astragal, right doors, on VHB
- 4× Pull, 96 centres, M4 × 10 button head

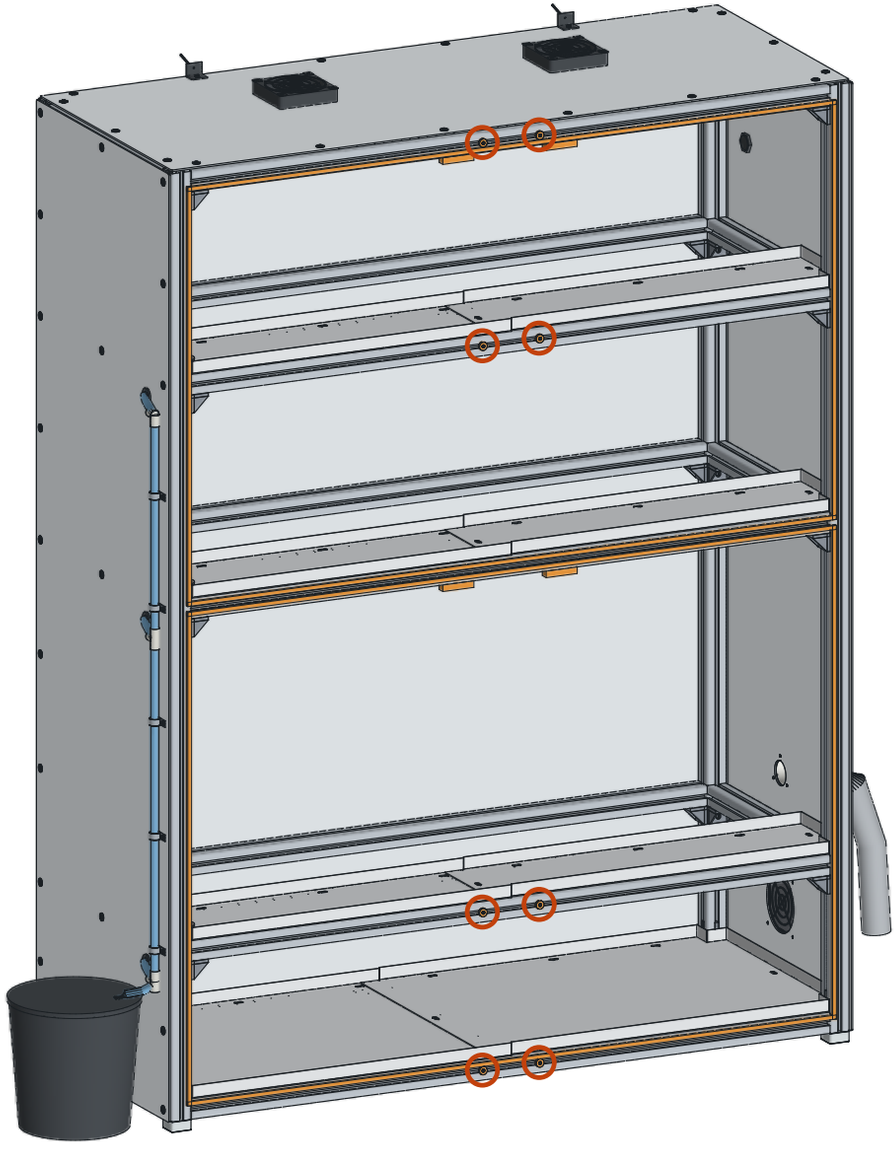
alcohol · 2.5 mm hex · a square

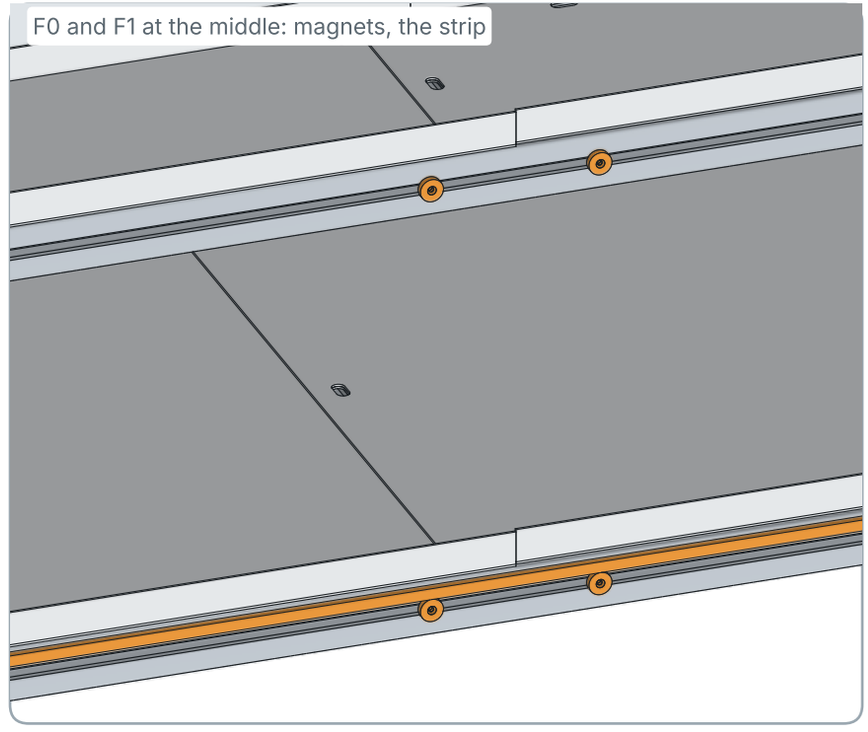
Drawn: the bottom right door, inner face up. The left doors mirror it, without the astragal.

Peel the inner film and wipe each bond spot with half alcohol, half water. Hinges: VHB on the door leaf, knuckle 0.5 off the hinge edge, square to it, at the heights on the front map; press hard for 30 s each. Discs where the magnets will meet them, 640 from the cabinet's left (mirror for right doors). Astragal: VHB on its outer 18.5, overhanging the right door's edge by 21.5 toward the left door. Reed magnet under its switch's spot. VHB reaches full strength in 72 h: hang the doors tomorrow.

15 Front · weatherstrip, magnets, switches

What the doors close on: strips on the lands, cup magnets on the slot lines, a switch under F2 and F4.





EPDM weatherstrip 5/16 × 1/4 in

8× Cup magnet Ø16 × 4.5, M3 × 12 + T-nut

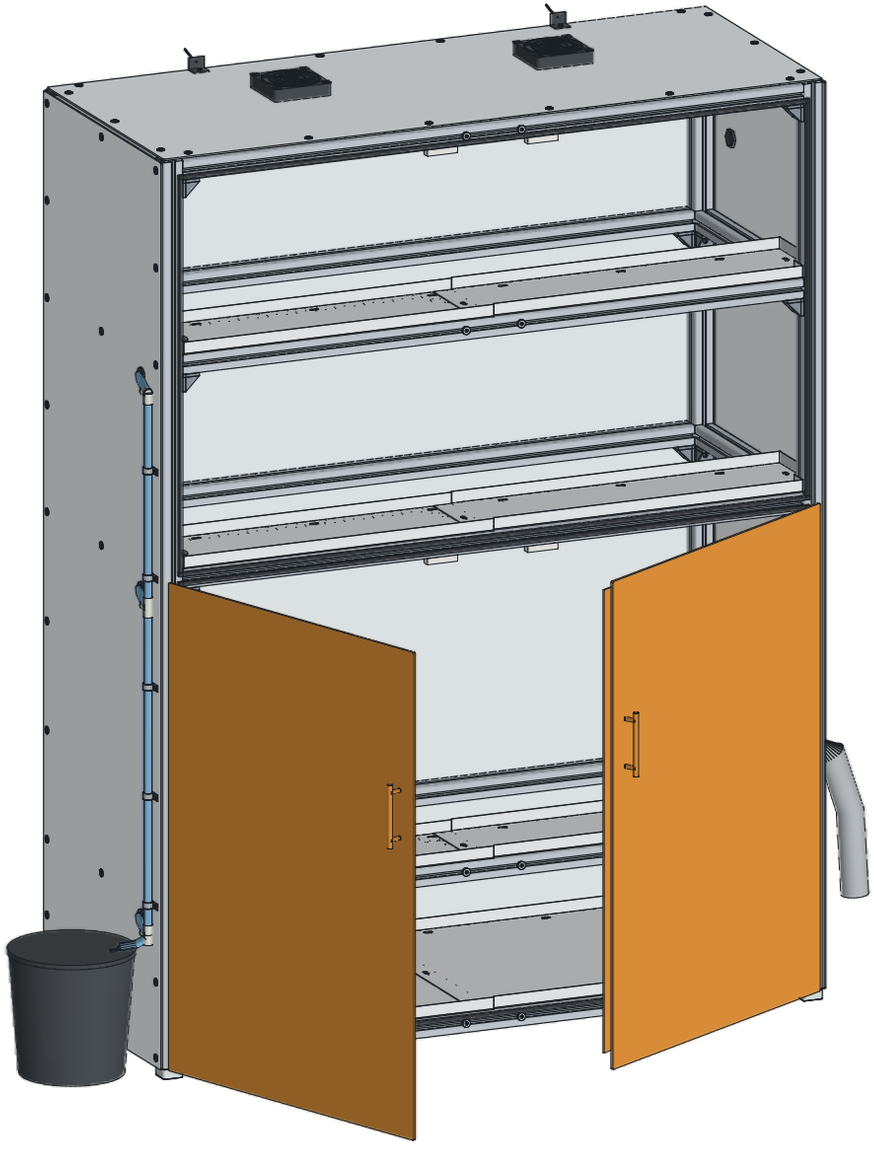
4× Reed switch 7939WG, under F2 and F4

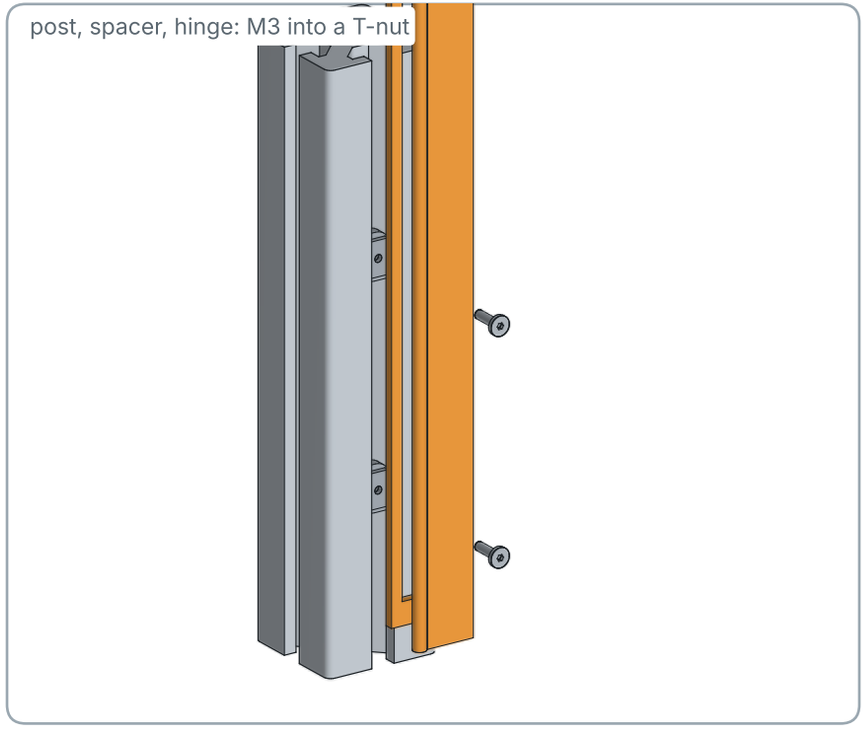
2 mm hex · knife

Strips: one loop for each pair, on the land beside each post's front slot and on the rails' lands: F0's upper, F2's lower and upper, F4's lower. Magnets on F0, F1, F3 and F4, 640 in from each side, face flush and tight. Switches on the underside of F2 and F4, 592 in from each side, on their adhesive; wire the four in series as the door chain (utility node, MCP23017 A4, wiring sheet E-03).

16 Front · hang the doors

Bottom pair first. Spacer on the post, the hinge's frame leaf on the spacer, one M3 in every hole.





8× Hinge spacer, slot on the slot line

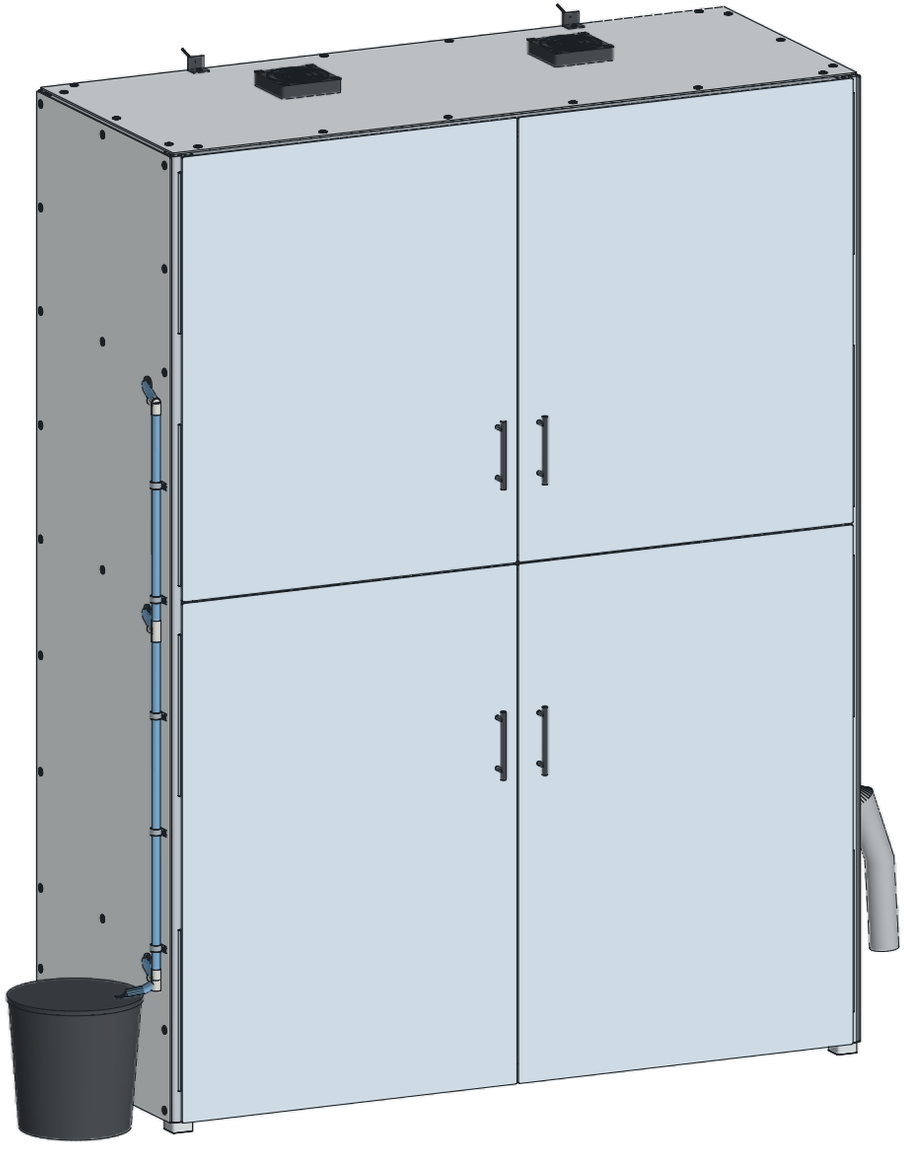
40+× M3 × 12 flat head + M3 T-nut, every hole

2 mm hex · a helper

Drop M3 T-nuts in the post's front slot, one per hinge hole. A helper holds the door open at 90°, hinges against the post; slip each spacer behind its frame leaf, slot over the slot line, and start every screw through the leaf, the spacer's slot and into its nut. Close the door gently: its free edge should meet both magnets together. Then the top pair, the same. Snug every screw once all four hang true.

17 Doors · adjust and check

Close each pair left door last. The gaps, the catch, the seal, the switch.



- [ ] **Centre gap**
  3 between each pair, top to bottom (2 to 5 is fine). Off? Re-seat the hinge in its spacer slot.
- [ ] **Split gap**
  3 between the pairs, even across. Off? A door's hinges sit at different heights.
- [ ] **Catch**
  Each door's free edge pulls in onto both magnets with a click; no door rests on its strip.
- [ ] **Seal**
  A strip of paper shut in the door anywhere round the loop drags as you pull it out.
- [ ] **Swing**
  Each door opens to 152° and touches nothing on the way.
- [ ] **Switches**
  The door chain reads open with any one door ajar, closed with all four shut.

Open order: left door, then right (the left door's edge lies over the right one's astragal). Close order: right, then left. If a door will not stay shut, look for something standing proud of the weatherstrip's plane: a screw head, a strip end lifted. Do not add magnets: find the high spot.

18 Before you plant

The checks that catch the usual faults, while the cabinet is still empty.

- [ ] **Every screw snug**
  112 M5s, the hinge M3s, the fan and grill M4s. A paint mark on each as you go.
- [ ] **Anchored**
  Both anti-tip cables taut, both wall brackets in studs. Pull the top forward: it does not move.
- [ ] **Air**
  Fans on: a tissue held at the intake filter is drawn in; nothing whistles at a door or seam.
- [ ] **Water**
  1 L into each tray. It drains through its grommet into the riser and the bucket; the panel stays dry.
- [ ] **Humidity**
  Humidifier on through the duct for 10 minutes: mist in tier A, no drips at the collar.
- [ ] **Cables**
  Everything in through the glands, the dome nuts tight on each bundle; no cable on a door's swing.

Clean the doors and panels with lukewarm water and a little dish soap, a soft cloth, no scraping. Isopropyl alcohol is fine for a spot; never glass cleaner (ammonia), acetone or thinners: they craze polycarbonate. The film comes off the outer faces last, once all the work is done.